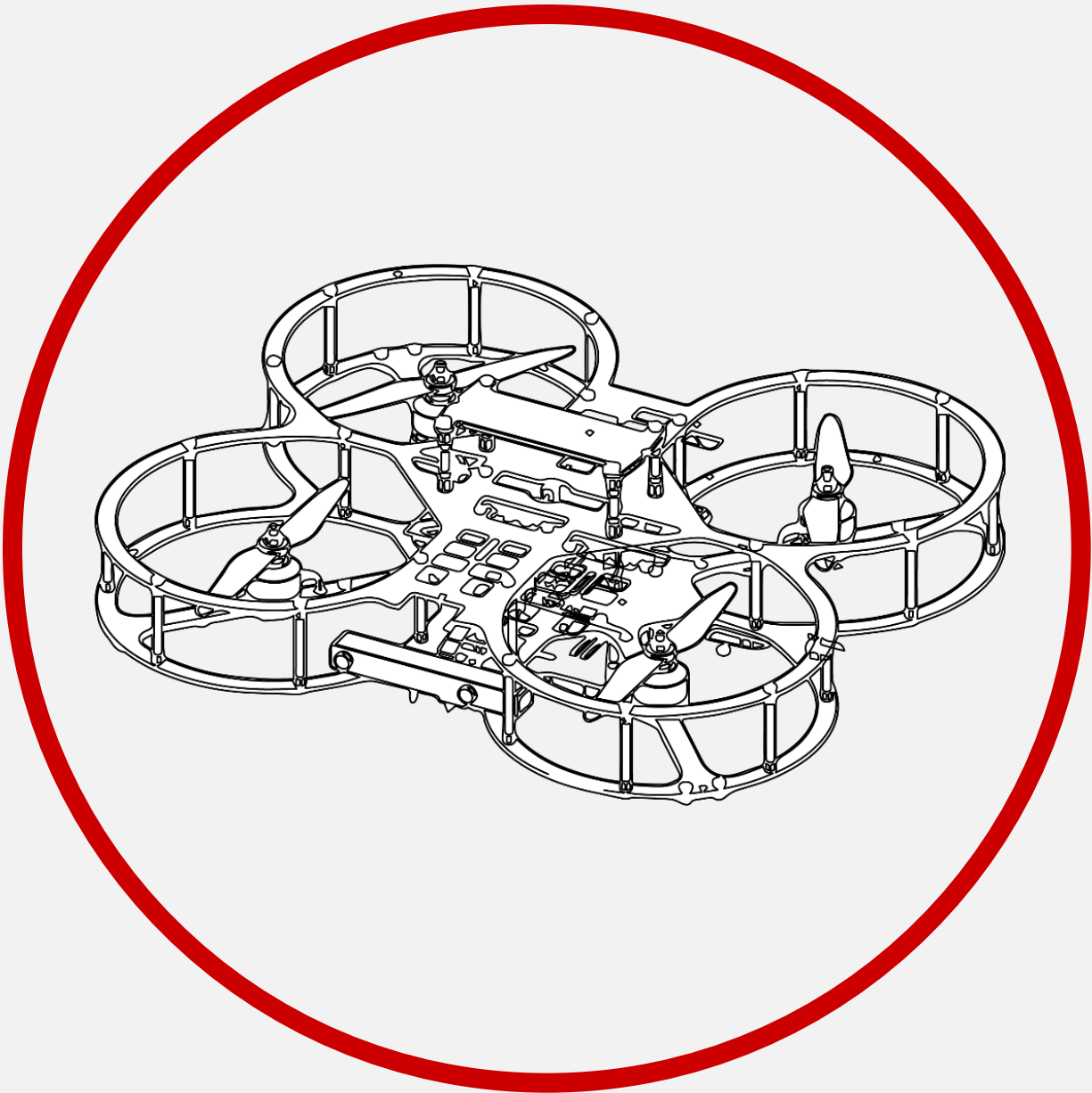




# Autonomous Vehicles Research Studio

Setup Guide – QDrone 1 Hover Test

*v 2.0 – 18th April 2023*

# Table of Contents

|    |                       |   |
|----|-----------------------|---|
| A. | Before the Hover Test | 2 |
| B. | QDrone 1 Hover Check  | 3 |

## A. Before the Hover Test

At this point, all the checkpoint tasks outlined in the table below should have been completed.

1. Your localization system should have been set up and functional.
2. You should be comfortable setting up and using the FrSky Taranis X9 Lite S Joystick
3. Your ground control station PC should be set up and functional.
4. The communication infrastructure between the ground control station PC, router and vehicles should be functional.

**Note:** Please contact the Quanser Engineer supporting you to start planning the commissioning and training process.

| Step | Done?                    | Section                                                               |
|------|--------------------------|-----------------------------------------------------------------------|
| 1/2  | <input type="checkbox"/> | Set up Workspace with Nets and Charge Vehicles                        |
| 3    | <input type="checkbox"/> | Mount OptiTrack Cameras                                               |
| 4    | <input type="checkbox"/> | Ground PC Setup (Connecting All Components)                           |
| 5    | <input type="checkbox"/> | Software Licensing and Testing (Sine Scope Demo)                      |
| 6    | <input type="checkbox"/> | Joystick Testing and Visualization Demo                               |
| 7/8  | <input type="checkbox"/> | Connectivity PC-Router-Vehicles (Vehicle Communication & TCP/IP Demo) |
| 9/10 | <input type="checkbox"/> | Camera Orientation and Calibration                                    |
| 11   | <input type="checkbox"/> | Rigid Body Definition (OptiTrack Visualization Demo)                  |
| 12   | <input type="checkbox"/> | Vehicle I/O Check                                                     |

The Hover Test for the drones is a test to ensure that all the components work together. The model will retrieve localization data from the Optitrack camera system. The user will issue commands via the joystick and a model running on the ground control station PC will forward the Optitrack camera data and joystick commands to a model running on the QDrone. The QDrone will be commanded to Arm, Takeoff, Land and Disarm, and will complete a Hover cycle in the process.

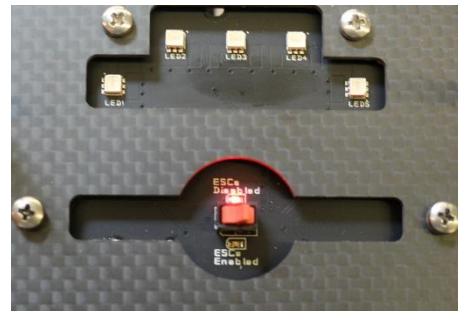
## B. QDrone 1 Hover Check

Ensure that the QDrone 1 IO Check has been run successfully before attempting to run this model.

1. Ensure that the Optitrack cameras have warmed up for 15-30 minutes. You can warm them up by keeping Motive open. If you have not calibrated the Optitrack cameras lately, do that first and define/export the Drone's rigid body before continuing.
2. Plug in and fasten a battery and turn ON the QDrone 1. Place it at the center of the workspace with the front camera pointed away from the ground PC. Ensure that the ESC disable switch is turned **OFF** (that is, the ESCs are **ENABLED**, which is indicated by a **green LED lit** next to the switch) (Figure 1a). Ensure netting is secured. Put on SAFETY GOGGLES.



a. ESCs Enabled (motors allowed to spin)



b. ESCs Disabled (motors can't spin)

Figure 1: ESC Disable switch in the ON and OFF position

3. From the same folder containing this file, open the folder QDrone Hover Test, and open Commander\_Stabilizer\_QDrone.mdl (Figure 2).

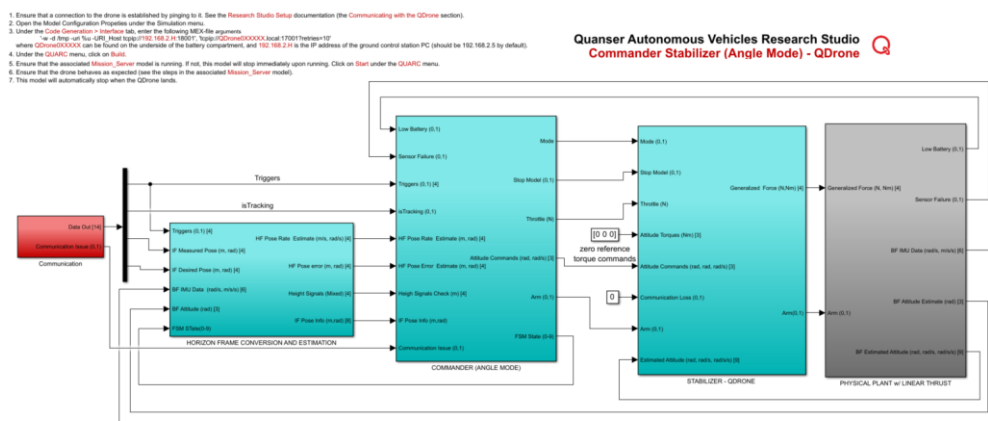


Figure 2. Commander Stabilizer Model

- ```
'-w -d /tmp -uri %u -URI_Host tcpip://192.168.2.H:18001', 'tcpip://QDrone0xxxxxx.local:17001?retries=10'
```

Configuration Parameters: Command\_Stabilizer\_QDrone/Linux64 Configuration (Active)

Search

Solver  
Data Import/Export  
Math and Data Types  
Diagnostics  
Hardware Implementation  
Model Referencing  
Simulation Target  
Code Generation  
Optimization  
Report  
Comments  
Identifiers  
Custom Code  
**Interface**  
QUARC

Code interface packaging: Nonreusable function

Data exchange interface

Array layout: Column-major

External functions compatibility for row-major code generation: error

Generate C API for:

☐ signals ☐ parameters ☐ states ☐ root-level I/O

☒ External mode

External mode configuration

Transport layer: quarc MEX-file name: quarc\_comm

MEX-file arguments: '-w -d /tmp -un %u -URI\_Host tcpip://192.168.2.H.18001\'tcpip://QDrone@xxxxx.local:17001?retines=10'

☐ Static memory allocation

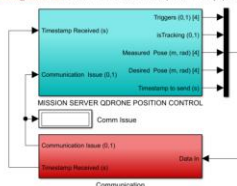
Deep learning

OK Cancel Help Apply

6. Ensure that a connection to the drone is established by pinging to it. See the vehicle communication document for more information.
7. Open a different MATLAB instance and from this window, in the same folder containing this file, open the folder QDrone Hover Test, and open Mission\_Server\_QDrone\_Hover\_Test\_FrSky.mdl (Figure 4). (If you have a Spektrum controller, open the file that does not end in FrSky).



- Place the **QDRONE** in the workspace and turn it ON (highlighted in blue).
- Ensure that the **ESC** display switch is OFF, i.e., the ESCs are enabled. This is indicated by a green LED next to the switch.
  - Ensure that the **MISSION** display switch is OFF, i.e., the **MISSION** display is disabled. The **MISSION** display, and ensure that the **Host Game Controller's** **Controller Number** parameter is set to 1.
  - NOTE: In case of unexpected behavior or emergencies, use the Emergency Stop push button on the Spectrum Joystick to cut power to the motors (this will result in the drone falling).
  - NOTE: In case of unexpected behavior or emergencies, use the Emergency Stop push button on the Spectrum Joystick to cut power to the motors (this will result in the drone falling).
  - Use the **QUADRC** mode, click on **Build**.
  - Click on **Start** then wait for the motor to start turning (wait ~1-5 seconds for the Motor API to stabilize and save location data).
  - In the **MISSION** SEVER **QDRONE** POSITION CONTROL sub-system, ensure that the **TrackingID** display shows 1 and that the **Joystick** **Issue** display shows 5 (Joystick must be powered ON and Controller must be powered ON).
  - NOTE: that the **Comm** issue with **Commander** display SHOULD display 1 under the **Commander**, **Stabilizer**, **QDRONE** mode is enabled. When it is running, this display should show 5.
  - NOTE: that the **Comm** issue with **Commander** display SHOULD display 1 under the **Commander**, **Stabilizer**, **QDRONE** mode is enabled. When it is running, this display should show 5.
  - Arm the drone by moving the Arm/Disarm toggle to the 2 position.
  - NOTE: that the **Comm** issue with **Commander** display SHOULD display 1 under the **Commander**, **Stabilizer**, **QDRONE** mode is enabled. When it is running, this display should show 5.
  - When finished, move the Takeoff/Land toggle to the 0 position to land the drone. The QDRONE should land.
  - NOTE: that the **Comm** issue with **Commander** display SHOULD display 1 under the **Commander**, **Stabilizer**, **QDRONE** mode is enabled. When it is running, this display should show 5.



4

8. Ensure that the joystick is turned **ON** by pressing the on button in the center for a couple seconds until the Quanser logo appears and make sure the **USB dongle is plugged** into the ground control station PC. If you have not used the joystick lately, unplug the dongle and plug it again to the PC to make sure the connection is working.
9. In the model that loads, under the blue **MISSION SERVER QDRONE HOVER** subsystem, open the green **JOYSTICK** subsystem (Figure 5a), and double-click on the **Host Game Controller** block (Figure 5b). Ensure that the **Controller number** (Figure 5c) drop down menu has selected the item labelled **FrSky Simulator**. (If you have the Spektrum controller and opened the file as per the instructions in step 7, the controller number should be set to Spektrum Receiver).

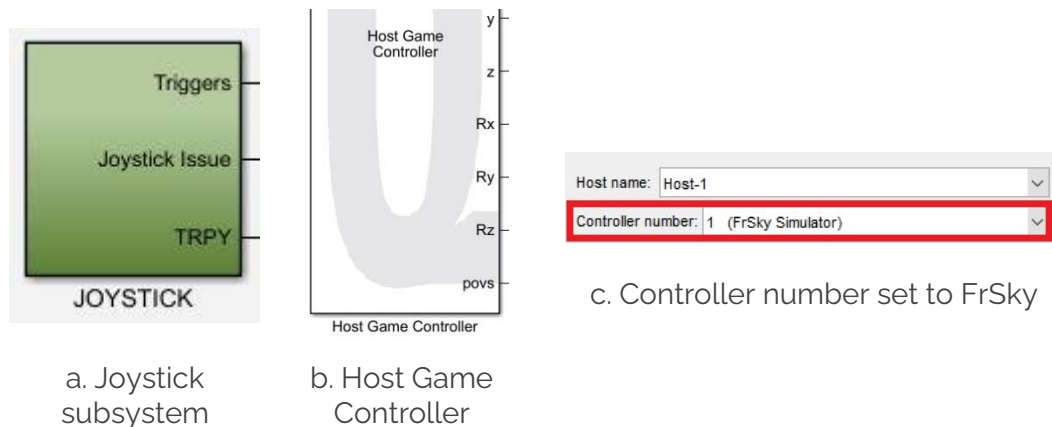


Figure 5: Selecting the FrSky receiver in Simulink/QUARC

10. Open the red **Data Server (Optitrack)** subsystem (Figure 6a) and double-click on the **Optitrack Trackables** block (Figure 6b). Select the calibration file (\*.cal) defined in **Calibrating Optitrack Cameras** file and the trackables definition file (\*.tra or \*.motive) defined in the **Rigid Bodies** file (Figure 6c).

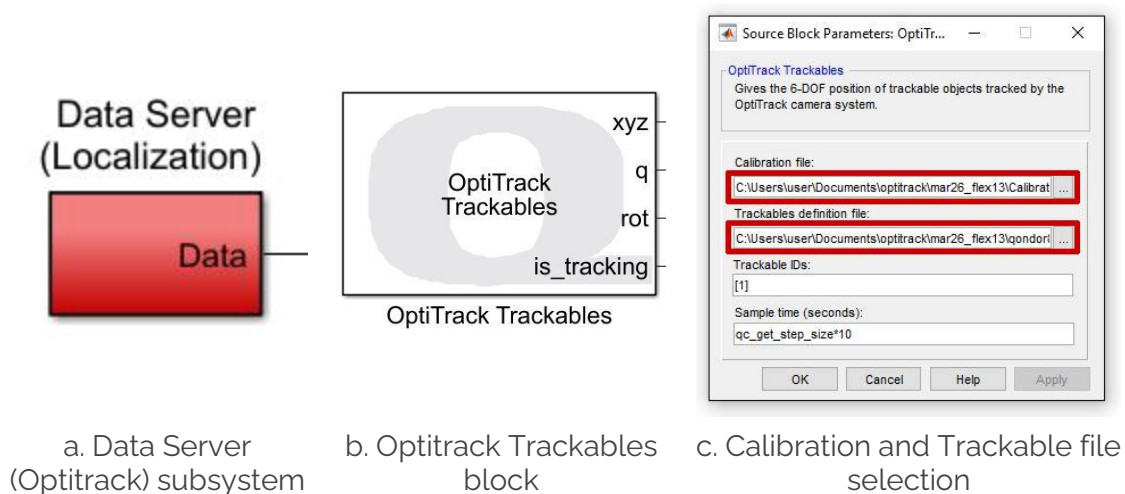


Figure 6: Specifying Calibration (\*.cal) and Trackable (\*.tra) definition files

11. On the Mission Server model, click on the **HARDWARE** tab on the top menu, and then click the green play button (**Monitor & Tune**). It should build and start the model. If you have an older version of Simulink that does not have a Hardware tab, under the **QUARC** drop down menu, click **Build** and once it finishes click **Start**.
12. Ensure the mission server starts running by checking the simulation time bar at the bottom right of the model (figure 7).



Figure 7: Simulation Time bar showing current time stamp

13. In the main diagram in the Localization Data Server area, ensure that the **isTracking?** Display reads 1. (Means drone is being tracked by localization system.)

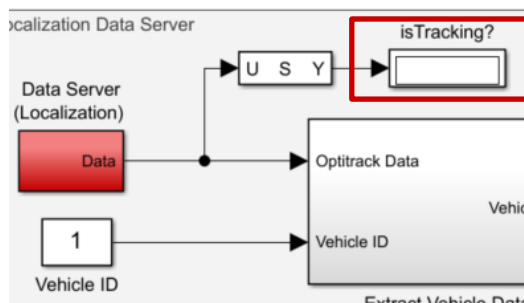


Figure 8. Is tracking display to confirm drone is being tracked

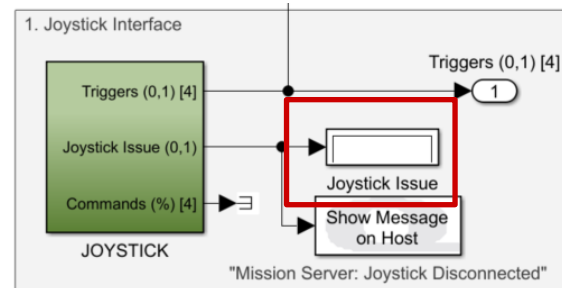
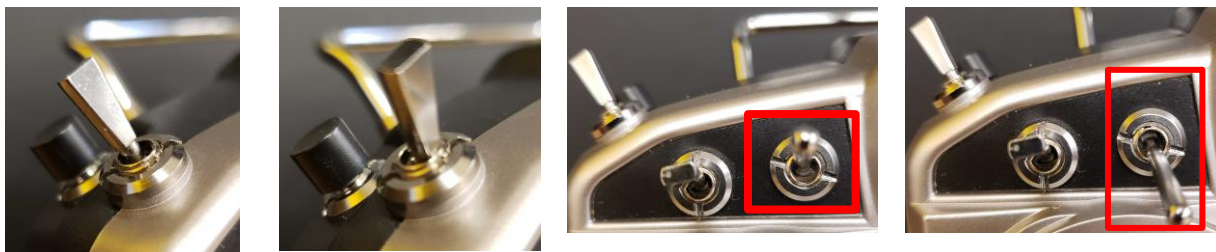


Figure 9. Joystick Issue display. A Zero (0) shows that there are no issues.

14. Ensure that the **Joystick Issue** display (Figure 9) reads 0 (no joystick issues). Ensure that all the toggles are set away from the user (see FrSky Joystick documentation). Make sure that the joystick's Arm/Disarm toggle is set as in Figure 10a and the Takeoff/Land toggle is set as in Figure 10c.



a. Disarm

b. Arm

c. Land

d. Takeoff

Figure 10: Joystick Arm/Disarm toggle (a and b) and LED toggle (c and d)

15. Follow step 11 again but switch to the **Commander\_Stabilizer** model and start it.



16. You should hear two beeps to signify that the ESCs are **ENABLED** and that the **Commander\_Stabilizer\_QDrone** model is running on the drone. The 5 LEDs on the top of the drone should turn **BLUE** (Figure 11a). This is the QDrone's **Disarmed** state.



a. Disarmed state of the QDrone 1



b. Armed state of the QDrone 1

Figure 11: Arm/Disarmed state of the QDrone 1.

17. After 2 seconds (initialization period), switch the **Arm/Disarm** toggle to 2 (towards the user) (Figure 10b). This should arm the drone and the motors should start spinning slowly. The 5 LEDs on the top of the drone should turn **GREEN** (Figure 11b). This is QDrone's **Armed** state.
18. Switch the **Takeoff/Land** toggle on the joystick to the 2 position (towards the user) (Figure 10d). The drone should takeoff and hover at the center of the workspace at the height of about 1m.
19. Make sure the drone can hover for a couple of minutes.
20. When ready to land the drone, move the **Takeoff/Land** toggle back to 0 (Figure 10c). The drone should slowly land at the center of the workspace.
21. Move the **Arm/Disarm** toggle to 0 (Figure 10a). The Commander\_Stabilizer\_QDrone model should automatically terminate. The Mission\_Server\_QDrone\_Hover\_Test model will throw an error since the Drone has been disconnected, this is expected. Stop that model.

This completes the **QDrone Hover check** task and confirms that your Autonomous Vehicles Research Studio is functioning correctly (including the Optitrack Localization System, Ground Control Station PC as well as QDrones). If you have any errors, make sure that all the steps prior to this checkpoint have been followed. If further issues persist, please contact Quanser technical support (tech@quanser.com).



© Quanser Inc., All rights reserved.



Solutions for teaching and research. Made in Canada.